Questions

Q1.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \qquad x > 0$$

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(a) Find $\frac{dy}{dx}$, giving each term in simplest form.

(2)

(b) Hence find the coordinates of the stationary point of C.

(4)

(c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C.

(2)

(Total for question = 8 marks)

(Q02 WMA12/01, Jan 2022)

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Q2.

A company makes a particular type of watch.

The annual profit made by the company from sales of these watches is modelled by the equation

$$P = 12x - x^{\frac{3}{2}} - 120$$

where P is the annual profit measured in thousands of pounds and £x is the selling price of the watch.

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According to this model,

(a) find, using calculus, the maximum possible annual profit.

(6)

(b) Justify, also using calculus, that the profit you have found is a maximum.

(2)

(Total for question = 8 marks)

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(**Q05 WMA12/01, June 2019**)

Q3.

A curve C has equation $y = 2x^2(x - 5)$

(a) Find, using calculus, the x coordinates of the stationary points of C.

(4)

(b) Hence find the values of x for which y is increasing.

(2)

(Total for question = 6 marks)

(**Q01 WMA12/01, Oct 2019**)

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Q4.

A curve C has equation

$$y = 4x^3 - 9x + \frac{k}{x} \qquad x > 0$$

where k is a constant.

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The point P with x coordinate $\frac{1}{2}$ lies on C.

Given that P is a stationary point of C,

(a) show that $k = -\frac{3}{2}$

(4)

(b) Determine the nature of the stationary point at P, justifying your answer.

(2)

The curve C has a second stationary point.

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(c) Using algebra, find the x coordinate of this second stationary point.

(4)

(Total for question = 10 marks)

(Q10 WMA12/01, Jan 2020)

Q5.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

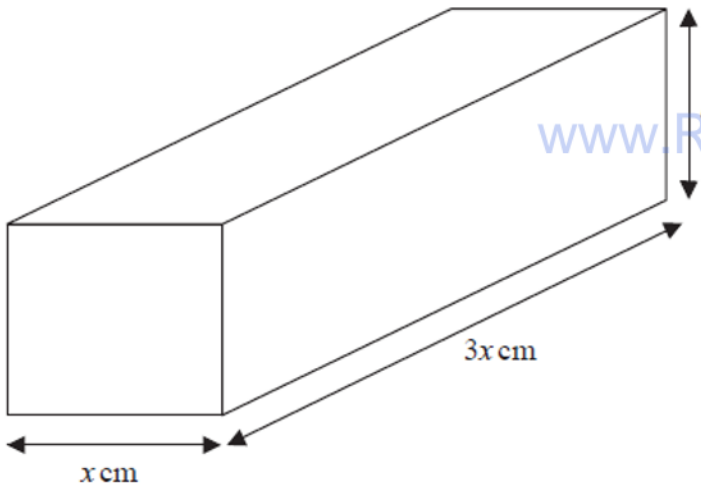


Figure 2

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A brick is in the shape of a cuboid with width x cm, length $3x$ cm and height h cm, as shown in Figure 2.

The volume of the brick is 972 cm^3

(a) Show that the surface area of the brick, S cm^2, is given by

$$S = 6x^2 + \frac{2592}{x}$$

(3)

(b) Find $\frac{dS}{dx}$

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(1)

(c) Hence find the value of x for which S is stationary.

(2)

(d) Find $\frac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S.

(2)

(e) Hence find the minimum surface area of the brick.

(1)

(Total for question = 9 marks)

(**Q02 WMA12/01, Jan 2023**)

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Q6.

The curve C has equation

$$y = \frac{12x^3(x-7)+14x(13x-15)}{21\sqrt{x}} \qquad x > 0$$

(a) Write the equation of C in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where a, b, c and d are fully simplified constants.

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(3)

The curve C has three turning points.

Using calculus,

(b) show that the x coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the x coordinate of one of the turning points is 1

(c) find, using algebra, the exact x coordinates of the other two turning points.

(Solutions based entirely on calculator technology are not acceptable.)

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(3)

(Total for question = 9 marks)

(**Q07 WMA12/01, Oct 2022**)

Q7.

A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

(a) Using calculus, find the x coordinates of the stationary points of the curve.

(4)

(b) Justify, by further calculus, the nature of all of the stationary points of the curve.

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(3)

(Total for question = 7 marks)

(**Q02 WMA12/01, Jan 2021**)

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Q8.

The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point (2, 5) lies on C
- the gradient of the curve at (2, 5) is 7

(a) find the value of a and the value of b.

(4)

(b) Prove that C has no turning points.

(3)

(Total for question = 7 marks)

(**Q10 WMA12/01, Oct 2019**)

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